5.4 Equilateral and Isosceles Triangles

Name _____ Period ____ Date _____

Geometry 5.4 · Student sheet

I can…

- use the Base Angles Theorem to find angle measures.
- use the converse to find side lengths.
- use the properties of equilateral triangles.

Vocabulary · fill in the missing word

Legs (of an isosceles triangle) — the two _____ sides

Vertex angle — the angle _____ by the legs

Base — the side _____ the vertex angle

Base angles — the two angles _____ to the base

Part 1 · Discovery: fold an isosceles triangle

Cut an isosceles triangle from a folded sheet. Measure.

Leg 1	Leg 2	Base angle 1	Base angle 2	Vertex angle

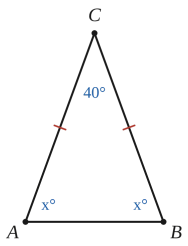
The angles opposite the congruent sides are _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Base Angles Theorem: if two sides of a triangle are congruent, then the angles _____ them are congruent.

Example 1

An isosceles triangle has a vertex angle of 40°. Find each base angle.



Converse of the Base Angles Theorem: if two angles of a triangle are congruent, then the sides opposite them are _____.

Example 2

Two angles of a triangle measure 50°, and the sides opposite them measure (3x − 2) and (x + 10). Find x and the length of each side.

Corollaries: a triangle is equilateral if and only if it is _____, so each angle of an equilateral triangle measures 60°.

Example 3

An equilateral triangle has sides of (4x − 3) and (2x + 9). Find x, the side length, and each angle.

Example 4

An isosceles triangle has a vertex angle of $(3x)°$ and base angles of $(x + 15)°$ each. Find all three angles.

Example 5

A roof truss is an isosceles triangle with base angles of $28°$. What is the angle at the peak?

Part 3 · Partner practice

Partner 1

An isosceles triangle has base angles of $48°$. Find the vertex angle.

Partner 2

An equilateral triangle has sides of $(5x - 8)$ and $(3x + 4)$. Find x and the side length.